

Topological Grid Segmentation - TGI Use Case Notes

[UC22-01] - Awesense Energy Transition Platform Use Case



Overview

This manual is intended to:

- Provide instructions to create the selected Use case in True Grid Intelligence (TGI).

For more details about Topological Grid Segmentation and how it can be analyzed using Awesense's Energy Transition Platform, please refer to the [UC22-01 - Topological Grid Segmentation.pdf](#) document.

TGI Use Case Notes

This document shows examples of using this Use Case along with a description of how to create such a Use Case on your data and screenshots directly from the TGI visualization interface.

Segmentations can be found directly in TGI in the main menu (select from the main menu - top hamburger icon).

True Grid Intelligence Segmentations

Search Grid...

300 km
200 mi

Segmentations

▼ Add Filter Search Table Create Segmentation

NAME	GRID	SEGMENTS	PUBLISHED	CREATED
transformer	North Central Zone	17,632	Yes	25 Apr, 2023 1:50 PM

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To create a new segmentation, click the "Create Segmentation" button.

Create Segmentation ✕

Select a grid and add segmentation rules to create a new segmentation.

Name:

test

Circuit to segment:

awefice

Description:

Enter description

RULES

Type:

--

is approximately

Value:

1

Weight:

1

Add Rule

Number of consumers

Number of transformers

Sum of kVA

Line length (km)

Create Segmentation

Parameters such as “Name”, and “Circuit” (e.g. North Central Zone or awefice) must be entered in the pop-up window. “Name” serves as the name of the segmentation, “Circuit” parameter denotes the ID of the grid which should be segmented.

Next, it is necessary to select by which criteria to segment the grid. For example, if we want to create a segment for each transformer, we enter in “RULES”, and “Type: Number of transformers” and enter 1 in “Value”. Weight is used when considering multiple criteria, so if using one criterion, enter 1. Then click on "Add Rule" and then "Create Segmentation".

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The progress of the segmentation is displayed in the upper right corner when you click on "Bell".

The screenshot shows the True Grid Intelligence Segmentations page. On the left, the 'Grid Element' details for 'arc:PRIMARY_CONDUCTOR' are displayed. The main map shows a region including Seattle, Portland, Boise, and Salt Lake City. An 'Activity Monitor' window is open in the top right, showing 'Segmenting grid awefice' with a progress bar at 51%. Below the map, the 'Segmentations' table lists the following data:

NAME	GRID	SEGMENTS	PUBLISHED	CREATED
transformer	North Central Zone	17,632	Yes	25 Apr, 2023 1:50 PM

Once the segmentation process has finished, the new segmentation will be displayed in the main menu on the Segmentations page. Individual segments can be selected and explored on the map or by clicking on "Base Segments" at the bottom left.

The screenshot shows the True Grid Intelligence Segmentations page with a detailed map of a city block. The 'test' segmentation is highlighted, showing 6 segments. The 'Segmentations' table lists the following data:

NAME	GRID	SEGMENTS
test	awefice	6
transformer	North Central Zone	17,632

At the bottom left, there is a button labeled 'Base Segments'.

Once the user is satisfied with the segmentation, the user can click “Publish Segmentation” and the segmentation instance will become available throughout the Awesense Platform.

General Note

This description was created for TGI version 3.10.0. For future versions of TGI, the process might be slightly different. Please contact tools@awesense.com for more information.